Total No. of Page - One *Roll No.*_____

Vth Semester

B.Tech.

Mid-Term Examination **September- 2023**

HU 307 Basic Econometrics

Time- 1.30 hrs **Marks- 25**

Note: 1. All questions are compulsory.
2. All questions carry equal marks.
3. Assume missing value, if any

1. What do you mean by Econometrics? Discuss its relevance for engineering students. [5][CO#3]

2. What do you mean by 'Residual' and 'Error' in OLS Model? Discuss. [5][CO#2]

3. Discuss significance of Gause –Markov theorem. [5][CO#1]

4. Discuss properties of an estimators? How to select estimators for a regression model? Discuss [3+2=5] [CO#1&3]

5. In a large manufacturing firm, an opinion survey was conducted regarding three type of bonus schemes. All employees were divided in four categories. The result is given below: [5][CO#2]

Employee Category	Bonus Schemes			Total
	Type I	Type II	Type III	
Labourer	190	243	197	630
Clerical	82	44	44	170
Technical	23	78	34	135
Executive	5	12	8	25
Total	300	377	283	960

Test the Hypothesis

H_0- Opinion about bonus schemes is independent of the types of employees.

$H_1 = H_0$ is not true

(The table value of chi-square for 6 d.f. and $\alpha = 0.05$ is 12.59)